

SOFTWARE PROBLEM STATEMENTS

HEALTHCARE

1. AI-Based Early Disease Prediction Using Minimal Symptoms

Problem Statement

Many individuals ignore early symptoms due to lack of awareness, leading to late diagnosis. Build a system that predicts potential diseases using minimal user input (2–5 symptoms).

Objective

Develop an AI-powered platform that:

- Takes basic symptoms as input
- Predicts possible diseases with probability
- Suggests next steps (doctor visit, tests)

Key Features

- Symptom input (text or voice)
- Machine learning model for disease prediction
- Confidence score for predictions
- Basic recommendations

Expected Output

- Web/mobile application with prediction dashboard
- Demonstration of model accuracy

Constraints

- Use publicly available datasets
- Ensure non-alarmist predictions

2. Mental Health Chatbot with Sentiment and Risk Detection

Problem Statement

Students and individuals hesitate to seek help for mental health issues. Build an AI chatbot that detects emotional state and risk level.

Objective

- Provide conversational support
- Detect stress, anxiety, or suicidal tendencies

Key Features

- NLP-based chatbot
- Sentiment analysis (positive, neutral, negative)
- Risk alerts for critical responses
- Suggestions for coping mechanisms and resources

Expected Output

- Interactive chatbot interface
- Sentiment analysis dashboard

Constraints

- Must include ethical disclaimers
- Avoid harmful or misleading advice